

12. References and Exact Instrument/Module Documentation

12.1 Standard Specifications & Manufacturer Technical References

1. **Intel Corporation**, *"ATX Specification Version 3 Multi Rail Desktop Platform Power Supply Design Guide, Revision 2.1a,"* Intel Engineering Design Center (EDC ID: 336521), 2023.
 - *Key Applied Clauses:* Section 3.2.1 (24-pin main power connector pin assignments), Section 3.2.2 (**PS_ON#** active-low control logic), Section 3.2.3 (**PWR_OK** power-good signal timing specs), and Section 3.2.4 (**+5VSB** standby rail specs).
2. **PLENTY COMPUTER Co., Ltd.**, *"ATX500WS Switching Power Supply Technical Specification & User Manual,"* Model ATX500WS, Rated Power 500W Max.
 - *Key Applied Parameters:* Nameplate voltage ratings, dual + 12V rail limits (+ 12V1: 13A, + 12V2: 14A), combined + 3.3V/ + 5V power limit (130W), and – 12V low-current rail limits (0.8A).
3. **International Electrotechnical Commission (IEC)**, *"IEC 60127-2: Miniature fuses - Part 2: Cartridge fuse-links,"* Fast-acting (F) glass tube fuses (5 × 20mm), 250V rated.
 - *Key Applied Parameters:* Time-current breaking characteristics (I^2t) for fast-acting branch wire protection.

12.2 Exact Instrument & Module Documentation Datasheets

1. XL6009 / LM2596 DC-DC Buck-Boost Adjustable Regulator Module

- **Manufacturer / Architecture:** XLSEMI / ON Semiconductor switching regulator IC architecture.
- **Input Voltage Range:** $3.5V - 28VDC$ (Operated from protected + 12V rail branch).
- **Output Voltage Range:** $1.25V - 26VDC$ continuously adjustable via multi-turn trimpots.
- **Current & Power Ratings:** Continuous Output Current: $1.0A - 3.0A$ (With mounted aluminum heatsink); Max Switching Frequency: $150kHz$ (LM2596) / $400kHz$ (XL6009).
- **Datasheet Document Reference:** *LM2596 3.0 A Step-Down Switching Regulator Datasheet (Rev. 26)*, ON Semiconductor Publication Order Number: LM2596/D.

2. DSN-VC288 Panel Dual Voltmeter & Ammeter Display Module

- **Module Identifier:** DSN-VC288 V1.0 Digital Panel Meter.
- **Operating Supply Voltage:** $4.5V - 24VDC$ (Powered from + 5V rail or + 12V rail).
- **Measurement Ranges:** Voltage: $0.0V - 100VDC$ (0.1% accuracy); Current: $0.00A - 9.99ADC$ (1.0% accuracy via internal $75mV$ shunt).
- **Display Specifications:** $0.28 - inch$ 3-digit dual 7-segment LED (Red for Voltage, Blue for Current); Refresh Rate: $\approx 300ms$.
- **Header Cable Definitions:**

- Thin Red: Instrument power input positive (+ 5V to + 12V)
- Thin Black: Instrument power supply common ground
- Thin Yellow: Voltage measurement sense input (0 – 100V)
- Thick Red: Current test path input (In series with load negative return)
- Thick Black: Current test path output to power supply ground

12.3 Laboratory Testing Equipment Specifications

- **Digital Multimeter (DMM):** UNI-T UT33D+ / Fluke 117 True-RMS Multimeter.
 - *Functions Used:* DC Voltage measurement (0.5%), Resistance measurement (0.8%), and Audible Continuity Testing.
- **Digital Storage Oscilloscope (DSO):** Rigol DS1054Z 50MHz 4-Channel Oscilloscope (Used for AC ripple and noise measurement with 20MHz bandwidth limit enabled).
- **Adjustable Electronic Load:** Atorch DL24 Electronic DC Load (150W max load capacity for rail voltage drop and load regulation testing).